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A Comparative Review on the Major Recourses and Technologies for Lithium Extraction

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Abstract

Lithium (Li) is the principal component of high-energy-density (rechargeable) batteries. From 2013-2023, Li demand has grown about sixfold and is expected to continuously rise due to the fast-growing global electric vehicle market. As part of this research, acid leaching from rock mining, pond evaporation of salar brines and direct lithium extraction from produced water are assessed as feasible Li extraction technologies. The first two technologies are commercially mature whereas the last one is largely under development. We review three main Li extraction technologies based on the published data and literature to investigate their pros and cons, application range/field and future prospect. Suitable types of source materials (e.g., solid vs. liquid), optimum and minimum required Li concentration levels, overall recovery efficiency, environmental impact, and prospects of future application are summarized and compared for each technology. This overall assessment of different technologies can be utilized to guide future R&D efforts on existing technologies or the development of new technologies. Conventional Li extraction method was initially designed and utilized for the mining of Li-rich mineral spodumene ($\text{Li}_2\text{O} \sim 6.0\text{--}7.5 \text{ wt.}\%$), but also applicable to other Li-bearing minerals such as petalite ($\text{Li}_2\text{O} \sim 3.5\text{--}4.5 \text{ wt.}\%$) and mica ($\text{Li}_2\text{O} \sim 2.0\text{--}7.7 \text{ wt.}\%$) with further adaptation. Due to the high recovery efficiency ($>90\%$), acid leaching of hard rock still plays a key role in stabilizing the Li supply chain. However, the high carbon footprint and harmful chemicals needed for extraction limit the overall benefit of this method on a long term. Pond evaporation of Li-enriched brines or salars (100s-1000s ppm) recently became a prominent extraction method in the Li supply market. This method is time and water consuming with significant land usage. Extensive Li reserves and a relatively lower carbon footprint make the technology promising for the future, especially if recovery efficiency can be improved. Direct Lithium Extraction (DLE) opens new avenues for utilizing low-Li concentration sources (10s-100s ppm), such as oilfield brines or geothermal water, which have not been considered in the past for being commercially viable. Although most DLE technologies are still in experimentation or pilot stage, these technologies are highly promising due to the estimated low carbon footprint and energy consumption for extraction. We provide a comprehensive review on the three main technologies mentioned, which are currently applied or under development for the extraction of lithium from different source materials. This assessment may be utilized to develop new technologies or improve existing technologies.

Introduction

Lithium (Li) is a critical metal that has been extensively used in modern energy storage technologies specifically as Li-ion batteries, and also as raw materials for many other products related to ceramics, glass, metallurgy, air treatment, pharmaceuticals, and polymers (Figure 1a-c). Lithium is the lightest metal and highly reactive and flammable. It is used as the principal component of high-energy-density (rechargeable) batteries. In the last decade, Li demand has grown more than sixfold and is expected to continuously rise dominantly driven by the global fast-growing electric vehicle markets (Figure 1). In addition, lithium-ion batteries are also ideal storage cells, with longer lifespan and higher power densities compared to other battery technologies, for storing and transporting "intermittent" electricity generated, for example by wind, tidal and solar, in a more cost-effective but less carbon emission way (Kesler et al., 2012; Kittner et al., 2017) that could support a more sustainable future. Lithium production dominantly relies on the extraction of natural geologic reserves such as lithium-rich hard rocks/clays and salt-lake brines, although other unconventional resources such as oilfield, geothermal, mine-water, and concentrated seawater brines may also be optimal (Bowell et al., 2020). However, geologic lithium deposits are commonly restricted in only a few geographic locations mostly within Chile, Australia, Argentina and China, which comprise > 77% of the worldwide Li geologic reserves (Figure 1d) and ~97% production (Figure 1e). Lacking access to economic Li reserves for most countries in the world will hinder large-scale production of Li and increase the risk of supply interruptions. For example, according to the European Union, lithium exceeded the threshold for economic importance and it is very close to the threshold of the supply risk (Alessia et al., 2021). Therefore, there are increasing interests to assess the global and regional accessibility of economic Li resources and the tailored technologies for Li recovery to stabilize the Li supply chain and reduce supply interruption risks. Below, we will review the three main Li extraction technologies by using different types of Li sources, including acid leaching from rock mining, pond evaporation of salar brines and direct lithium extraction from produced water. Although some technologies are considered economically mature, challenges are still identified in these technologies that require further investigation and improvement.

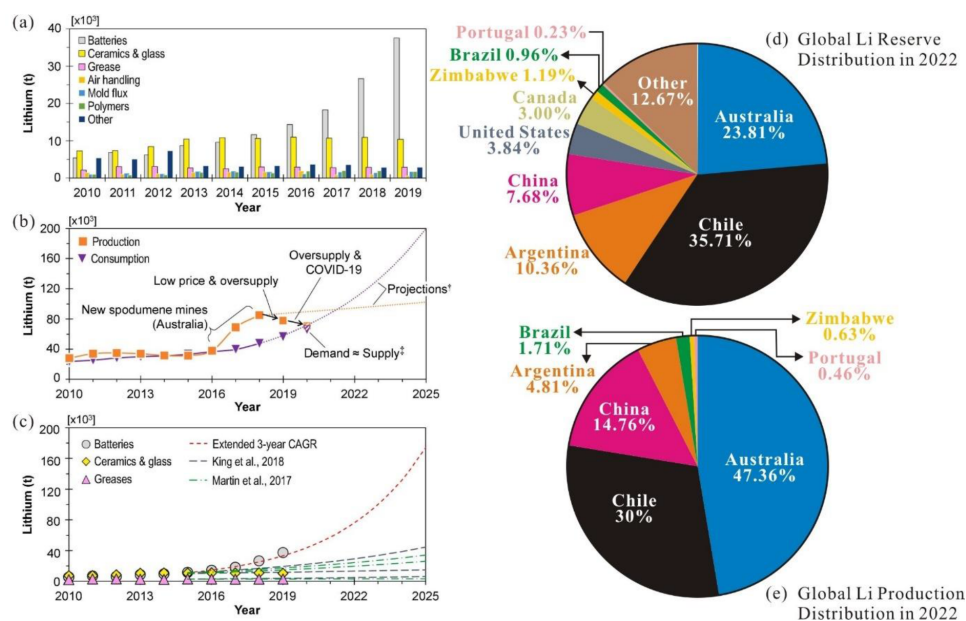


Figure 1—Summary of global lithium demand and supply showing (a) breakdowns of the key lithium applications from 2010 to 2019, with a significant increase in batteries, (b) lithium production vs. consumption from 2010 to 2019, with forecasts (indicated by dashed lines) until 2025, and (c) global Li demand of the three main Li consumers - battery, ceramics & glass, and grease - and projections of their growth until 2025 from Tabelin et al. (2021), together with global distribution of lithium reserves and production in 2022 by different countries (data from 2023 Lithium Industry Worldwide Report, Statista).

Comparative Review of the Three Main Lithium Extraction Technologies

Hard Rock Extraction Method: Acid Leaching

Although the advancement of new extraction technologies such as Direct Lithium Extraction (DLE) may in the future completely change the lithium supply chain (e.g., Vera et al., 2023; Farahbakhsh et al., 2024), hard rock mining (also known as open-pit mining) followed by acid leaching processes still plays a vital role today as it is the most time-efficient way for large-scale Li production. The conventional method for lithium extraction is primarily applied to ore-grade Li-rich rocks, specifically spodumene, and includes a series of mechanical and chemical processes (Ellestad and Milne, 1950) that have changed little in the past century (e.g., see a review in Gao et al., 2023; Krishnan and Gopan, 2024). A similar procedure is also used for Li-bearing clay minerals such as lepidolite, zinnwaldite, masutomilite, swinefordite, hectorite, cookeite and jadarite, but further improvement is still needed to increase the recovery efficiency for industry-scale production (Zhao et al., 2023).

As shown in Figure 2, mined hard rocks, such as spodumene ($\text{LiAlSi}_2\text{O}_6$), a Li-enriched aluminosilicate mineral, are first crushed and grinded to liberate individual mineral grains and increase their surface areas. Then thermal treatment at ~ 1000 – 1100°C will be applied to convert the naturally occurring, less reactive α -spodumene to metastable, more reactive β -spodumene through phase transformation. β -spodumene will be exposed to concentrated sulfuric acid at ~ 200 – 250°C to convert lithium into soluble lithium sulfate (i.e., acid leaching). The roasted materials will be further washed with water to dissolve lithium sulfate, separating it from insoluble silica and other residues. Finally, lithium will be recovered, commonly as Li_2CO_3 through precipitation or other lithium compounds via additional chemical processes, followed by further filtration and purification.

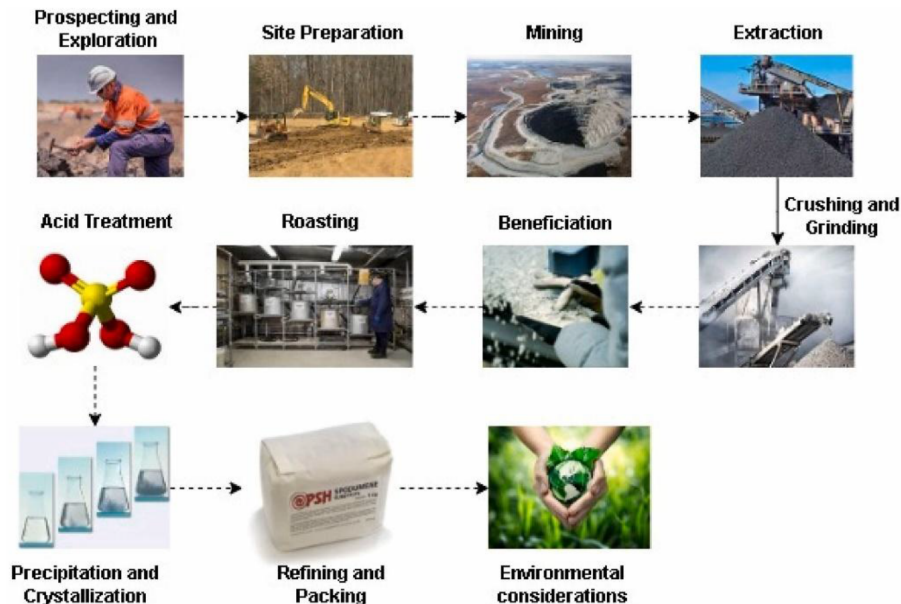


Figure 2—General physical and chemical processes of the conventional extraction method for the recovery of Li from spodumene (from Krishnan and Gopan, 2024).

However, this conventional Li extraction method is only commercially applicable to processing of spodumene ($\text{LiAlSi}_2\text{O}_6$, a Li-enriched aluminosilicate mineral), although other Li-bearing rocks containing >1.0 wt.% Li_2O may also be suitable (e.g., Yelatontsev and Mukhachev, 2021). The general processes included in the conventional Li extraction approach are summarized in Figure 2, and also described in detail in Yelatontsev and Mukhachev (2021), Zhao et al. (2023) and Krishnan and Gopan (2024). Spodumene and other Li-bearing rocks are commonly mined by the open pit method, followed by crushing and

milling of the mined rock blocks to liberate individual mineral grains and increase their surface areas. The crushed minerals are then sent to a flotation cell for separation of target ore minerals such as spodumene. The separated spodumene is roasted to convert the naturally occurring, less reactive α -spodumene to metastable, more reactive β -spodumene through phase transformation at the temperature range of 920-1050 °C (e.g., [Tkachev et al., 2019](#)) depending on the roasting conditions, presence of impurities, and some other factors. After roasting, the converted spodumene is further exposed to concentrated sulfuric acid under the temperature range of 180-300 °C to selectively leach out Li as Li_2SiO_4 and minimize the extraction of impurities (e.g., [Samoilov et al., 2018](#); [Wang et al., 2020](#)). After further purification, Li can finally be precipitated as lithium carbonate (Li_2CO_3) by adding soda and/or lime (e.g., [Samoilov and Kulenova, 2008](#); [dos Santos et al., 2019](#)). Other final Li products such as lithium hydroxide (LiOH) and lithium oxide (Li_2O) can be also obtained if the rock processing method is tailored to the products (e.g., [Yelatontsev and Mukhachev, 2021](#)).

As noted in [Haddad et al. \(2023\)](#), the chemical processing of lithium consumes extensive energy and water (e.g., ~60 mwh energy and ~70 m³ water for producing 1 tonne of lithium salts). Also depending on the feedstock, producing 1 tonne of Li_2CO_3 or LiOH releases between 3 and 17 tonnes of carbon dioxide. As shown below in [Figure 3](#), processing accounts for ~70% of CO₂ emissions associated with lithium production, whereas mining and transport are responsible for the rest. It is further noted that Li_2CO_3 production generated 3.6 million tonnes of CO₂ in 2020, and by 2050, the IEA data suggests those emissions could grow to 60 million tonnes of CO₂ per year.

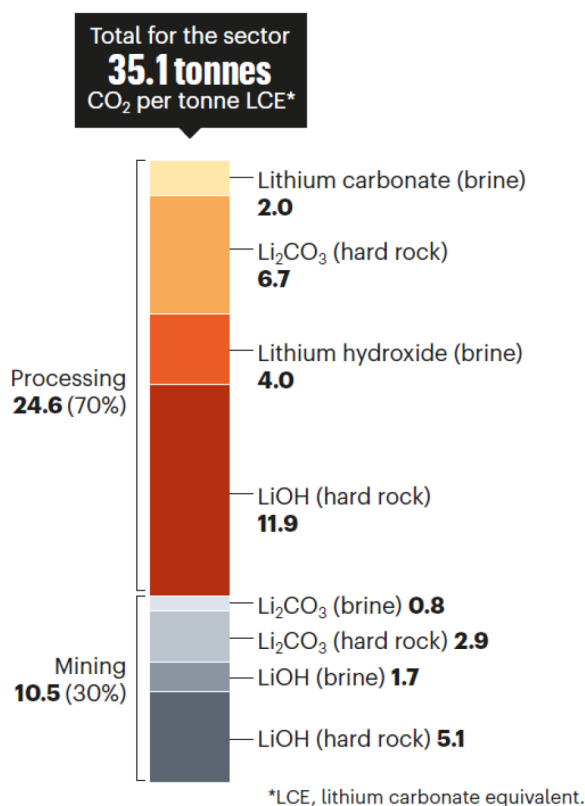


Figure 3—Total emissions estimated for the Li extraction. Chemical processing of lithium ores, clays and brines releases 70% of emissions from the sector, owing to the use of heat and carbon-intensive reagents, such as calcium hydroxide, sodium hydroxide and sodium carbonate (from [Haddad et al., 2023](#)).

Salt Lake Brine Extraction Method: Pond Evaporation and Precipitation

Due to the rapid exploitation and very limited amounts of deposits, reserves of Li-bearing ores are diminishing quickly, necessitating alternate Li supplies to stabilize the market. Salt-lake brine or salar brine,

on the other hand, is one of the most potential lithium sources in nowadays as almost 80% of total land-based lithium reserves globally are salt-lake brines (Zhang et al., 2019). Among these sources, continental brines, mainly from Argentina, Bolivia, and Chile in South America, account for the largest amount of lithium resources in the world (Figure 1e). Although many technologies for lithium extraction from brines have been investigated in the past few decades, evaporation followed by subsequent precipitation and purification method is still the most optimal way due to several advantages, including lower cost, easier industrialization, and lower overall carbon footprints (Heidari and Momeni, 2017). This method is a multi-stage, solar-driven process that concentrates lithium in the brine followed by Li precipitation, commonly as lithium carbonate (Li_2CO_3). As shown in Figure 4 below, Li-rich brine, typically containing relatively lower Li concentrations at 100-1000s ppm level, along with high levels of other interfering elements such as Ca^{2+} , Mg^{2+} , Na^+ and K^+ , is first pumped from underground aquifers and transferred to a network of evaporations ponds. Then using solar energy, lithium will be gradually concentrated within the brines by evaporating water. To expedite this process, brines may be moved sequentially through a series of ponds, during which different salts precipitate and crystallize out due to the changing concentrations of different elements. Depending on the climate, total residence time can vary from month to year. Final lithium precipitation, usually as lithium carbonate, is achieved using sodium carbonate. The crude lithium carbonate is separated from the solution using filtration, followed by washing with deionized water to help remove other adhering impurities such as NaCl and KCl.

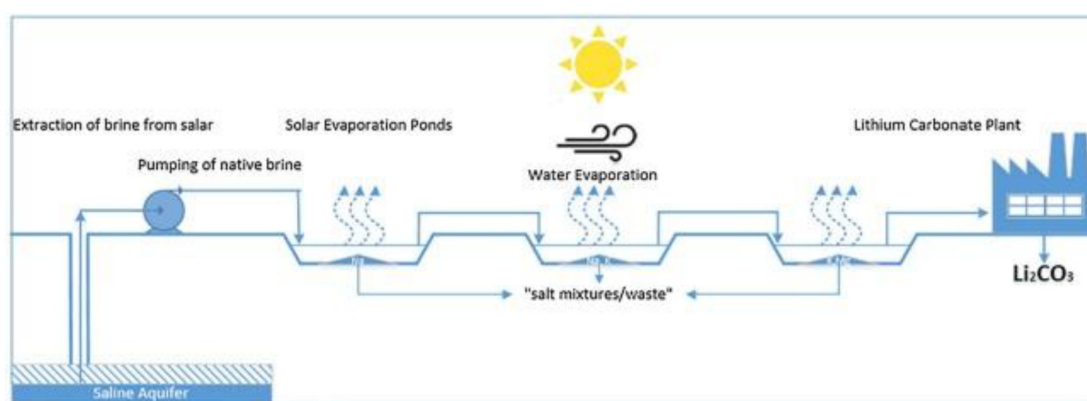


Figure 4—Generalized workflow for pond evaporation and Li precipitation from brines (from Flexer et al., 2018)

Unconventional Source Extraction Method: Direct Lithium Extraction from Produced Water

Direct Lithium Extraction (DLE) from produced water represents the latest innovative and potentially transformative approach to expanding global lithium supply (e.g., Kumar et al., 2019; Nikkhah et al., 2024; Wang et al., 2024). Unlike traditional hard rock or brine sources, produced water is often readily available, continuously generated and abundant during production. Because the produced water requires further process anyway, its use for lithium recovery aligns with both water optimization and environmental remediation goals.

Certain oil and gas field brines around the world contain high concentrations of Li and some of them even comparable with the salar brines (Figure 5). For example, Smackover brines in the U.S. have a maximum of over 500 mg/L of Li, showing great potential for further processing and recovering lithium from the brines. Furthermore, DLE from oil and gas produced waters on site can take advantages of the existing production wells and field infrastructure and benefit from the revenue stream generated by Li recovery from the wastewater, which would otherwise be a financial burden.

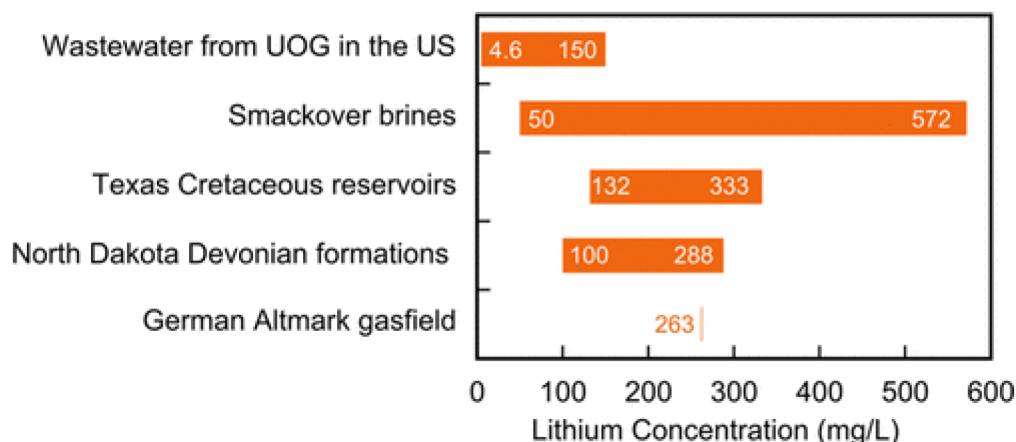


Figure 5—Lithium concentrations from U.S. oil and gas fields (from Kumar et al., 2019)

Several promising pilot projects and R&D studies have shown that DLE technologies can selectively extract lithium ions from produced water with very low lithium concentrations and accompanied by high levels of interfering ions such as Na^+ , Ca^{2+} and Mg^{2+} . However, challenges still remain as (1) the low lithium concentrations require highly selective and robust extraction system that may need extensive initial capital investment, (2) depending on the geological formation and production method, the chemical compositions of produced water can be highly variable and hence, requires site-specific tuning of DLE processes, and (3) operational integration with oil and gas infrastructure is complex, resulting in highly uncertain economic viability at commercial scale.

Table 1

Technology	Resources	Recovery Rate	Time Scale	Technique Maturity	Pros	Cons
Acid Leaching	Hard Rock with high Li concentration	>90%	Weeks	Commercial	High yield, well-established process	Energy-intensive, acidic waste
Pond Evaporation	Salr brine with high to moderate Li concentration	40-60%	Moths to years	Commercial	Low cost, simple design	Long duration, low recovery, geographic limitation
DLE	Oilfield brine, geothermal with low Li concentration	>80% (potential)	Hours to days	Pilot-Early Stage	Fast, more sustainable	Technology readiness, cost

Conclusion

The global shift toward renewable energy has significantly increased the demand for efficient, scalable and more sustainable lithium sources and extraction technologies. We have compared three primary methods currently in use or under development including acid leaching from rock mining, pond evaporation of salar brines and direct lithium extraction from produced water. As summarized in Table 1, acid leaching of spodumene or other Li-rich rocks is a well-established and high-yield process. It benefits from mature infrastructure and high Li recovery and efficiency, but suffers from high energy consumption, environmental hazards related to strong acid usage and the need for extensive calcination. In contrast, pond evaporation leverages low-cost solar energy and is economically viable in arid climates with high evaporation rates. However, it requires long processing times and suffers from lower recovery efficiency and significant land and water usage, which raises significant social and environmental concerns. DLE represents a

transformative innovation with the potential to radically improve lithium recovery, reduce processing time and minimize the water and land usage. It also opens new avenues for utilizing low-Li concentration sources (10s-100s ppm), such as oilfield brines or geothermal water, which have not been considered in the past for being commercially viable. While numerous DLE technologies are being piloted worldwide, commercial-scale implementation remains limited due to technical complexity, brine property and high initial capital investment. In the future, lithium extraction will likely involve a combination and diversification of technologies to address resource variability and environmental and social constraints.

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